My research is committed to developing trustworthy and equitable ML solutions that support accessible and scalable healthcare interventions across diverse settings. By fostering trust and promoting a deeper understanding of ML technologies among healthcare professionals, we can bridge the gap between cutting-edge advancements and their practical applications. Extending these technologies equitably can empower diverse patient populations with greater health insights, enabling more informed decision-making through inclusive healthcare experiences. A PhD at VUMC DBMI will provide me with the expertise and mentorship necessary to address healthcare inequities in democratizing predictive analytics.

**Computation and Health:** Under the guidance of Dr. Bradley Malin at Vanderbilt University, I explored the privacy-utility tradeoffs of the novel distributed learning framework, split learning (SL), through model inversion attacks at the cut layer. I am particularly curious about SL as a potentially superior alternative to federated learning in high-cost settings like healthcare, prioritizing client needs with reduced computational costs, enhanced model privacy, and flexibility to support heterogeneous hyperparameters and data. In addition to readmission prediction and data privacy, my findings encouraged me to think about the ethical implications of bias and accessibility that motivate distributed learning. Notably, I am interested in how we can leverage distributed learning to address heterogeneity in patient demographics, computational resources, and image acquisition protocols. One pathway towards this is interpreting latent representations at the cut layer in SL, which could offer privacy and utility insights into systematically optimizing cut layer configurations for diverse tasks.

After exploring institution-level privacy-preserving collaboration, I investigated ML's potential to directly influence clinical decision-making. At USC, I received the CURVE Fellowship to investigate saliency methods in biological brain age (BA) predictions under the guidance of Dr. Andrei Irimia. Deep neural networks can estimate BA from structural features found in magnetic resonance imaging, demonstrating strong potential as an early identification tool for neurodegenerative disease. My project compared saliency methods' capacities for neuroanatomic insight to foster clinician trust toward black-box BA predictions. The Integrated Gradients saliency method yielded superior neuroanatomic insights; however, I was more intrigued by the Input x Gradient method, which achieved comparable results with significantly lower computational demands. This discovery could support more reliable and interpretable integration of modern ML technologies in low-resource institutions, allowing them to harness technologies historically out of reach and even contribute their own research findings. Empowering previously underrepresented healthcare institutions with the tools to leverage and better understand cutting-edge advancements is the first step towards combating perpetuated health inequities. My fellowship culminated in a presentation at the CURVE research symposium and led to my summer role as a full-time research assistant, where I first-authored our *Neuroinformatics* publication. My current research explores aging and cognitive decline using local (voxel-level) BA across social determinants of health and cardiovascular conditions.

**Other Computational Research:** Before diving into health-focused research, I participated in projects focused on education and accessibility under Dr. Yu Sun. My first project utilized k-Means clustering and Mediapipe to analyze paired videos of a student and coach performing the same action like a golf swing, tennis stroke, or gait exercise. Additionally, I implemented a transformer-based sentiment analysis framework to streamline dense academic texts into

digestible outlines, allowing young students to engage with material that previously felt inaccessible. These initial projects introduced me to the versatility of AI in creating socially impactful technologies to serve communities I am involved in and care about.

After these initial experiences, I sought out larger NLP focused opportunities and joined the NASA Aeronautics Research Institute's Advanced (NLP) Methods team. Because domain-specific speech contains phraseology not encountered in typical English, existing NLP tools that solely rely on transformers and/or other neural architectures struggle to properly format speech transcripts in aeronautics. To address this challenge, I proposed a hybrid rule-based/neural inverse text normalization framework. In my four semesters with Ames, I led the project's direction, timeline, and goals, resulting in a first-author publication and presentation at AIAA 2024. In the coming years, this project will be deployed into the Federal Aviation Administration Command Center to digitize and automate National Airspace webinar review. While my focus has shifted from aeronautics, the experience ideating, developing, and deploying this large-scale project equipped me with a foundation for tackling novel challenges with interdisciplinary methods—insights I apply to my current research.

**Future Research:** Recent advances in data-gathering, wearable technologies, and predictive analytics have transformed healthcare; however, these tools remain out of reach for many communities. My research hopes to advance healthcare ML that is not just accurate in theory, but inherently accessible and generalizable across diverse applications. I believe in achieving this goal on three fronts: (1) rigorous validation and interpretation of predictive technologies, (2) frameworks that extrapolate across heterogeneous populations and sites, and (3) secure collaboration across community clinics and large institutions. Rigorous validation ensures that emerging tools are as transparent as they are accurate, enabling clinicians to make fully-informed decisions tailored towards each patient's specific needs. It is equally important to ensure that distributed learning is secure and that frameworks account for variance in patient demographics, healthcare environments, and resource availability.

**Why VUMC DBMI:** VUMC DBMI's strong record of cultivating leading academics will prepare me to lead an academic research lab and mentor the next generation in trustworthy and equitable healthcare ML. My NSF REU at Vanderbilt immersed me in a collaborative community of interdisciplinary thinkers from backgrounds across policy, medicine, and computation, located in a healthcare hub that I hope to work in someday. Vanderbilt's premier medical system hosts readily available clinical datasets, and collaborating physicians and sites to deploy advancements in. I would love to continue studying under Dr. Bradley Malin, exploring how advancements in data privacy and fairness can support secure and efficient collaboration in healthcare settings. I'm also interested in diving further into explainable AI and interpretable ML for clinical decision support under Dr. Siru Liu.

**PERSONAL STATEMENT**

My journey is shaped by personal and family health challenges that have instilled resilience and a commitment to advancing equity in education and healthcare. During the COVID-19 pandemic, I organized an afterschool tutoring program to support peers struggling with the shift to remote learning. Tutoring became a crucial way to support my family after [REDACTED FOR PRIVACY REASONS—INSERT SOME PERSONAL CHALLENGES FOR CONSISTENCY]. Just two months after committing to Purdue University, ready to take my first steps toward an eventual career in academia, I decided to remain home to spend time with and care for my mother. We both struggled to accept the situation, but ultimately, I believed in myself to get back on track, my mother had shown me just how to do so.

My first year at community college was not the fresh start I had hoped for; While my friends pursued clubs and research labs at their new schools, I searched online job boards. I found my first teaching role at Coding Minds Academy, a K-12 computer science school. Within months, I expanded from teaching to leading workshops, training employees, and developing project curricula. My efforts impressed Professor Yu Sun, the academy's founder, who promoted me to project lead for the advanced track students and assessment instructor for incoming students. Professor Sun opened my eyes to the impact of a truly inspiring mentor—not only had he guided me into formal teaching and research experiences, but he understood and accommodated my personal situation with extra flexibility and more classes to teach when finances were tough. This experience inspired me to pursue further research like the NASA Community College Scholars and L'SPACE programs, and eventually an AI/ML Research Intern position with NASA Ames.

When my mother's condition stabilized, I transferred to USC, where I faced my own physical challenges, including bilateral hip and shoulder labrum tears. These injuries made daily tasks challenging and forced me to juggle frequent hospital visits and intensive physical therapy sessions alongside academic and financial commitments. My father unemployed and my mother on reduced hours, I still felt responsible to support them while staying true to my own academic goals. Through stringent time-management and discipline, I balanced a 20-hour-per-week NASA internship, 10-hour-per-week research fellowship, and leadership positions in HackSC—all while taking three to four computer science courses per semester to catch up as a transfer student.

After my NSF REU at Vanderbilt, I had hoped to continue my research in ethical AI and privacy-preserving distributed learning under Dr. Bradley Malin; however, volunteering for the position was not financially feasible. Sacrificing this opportunity underscored the importance of flexibility and perseverance, as I then secured an on-campus fellowship under Dr. Andrei Irimia, who provided flexible support and pushed me to achieve a second publication.

My journey has solidified a commitment to fostering inclusive and supportive learning environments. After transferring to USC, I remained in contact with Dr. Sun, through whom I connected with and guided three students towards successful transfers into USC. As a lead in HackSC, I co-organized initiatives like HackSC Jr. in partnership with the National Society of Black Engineers, introducing underrepresented groups to computer science. This year, I helped launch USC's first ethics bowl team to promote discussions on the ethical implications of technology. My values in education and accessibility are shaped by the personal adversity I've overcome and the mentorship I've been fortunate enough to receive—values that I hope to once again bring to a diverse campus and program like VUMC DBMI.